Quantitative Macro Theory **HW #4**
Spring 2019 **Due at 5pm on *Sunday* 2/24/2019**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (100 points) **Income-Fluctuation problem**

   Consider the problem

$$V(a, y_i) = \max_{c,a'} u(c) + \rho\beta \sum_{j=1}^{3} \Gamma_{i,j} V(a', y_j)$$

   subject to

$$c + a' = Ra + y_i,\ a' \geq \underline{a}.$$

   Calibration:

   - The survival probability is $\rho = 0.975$.
   - $u(c) = \frac{c^{1-\sigma}}{(1-\sigma)}$, with $\sigma = 2$.
   - The discount rate, $\beta = 0.93$.
   - Gross interest rate, $R = 1.025$.
   - Income is given by y. Use a 3-state Markov chain Γ that we loosely identify with "super-rich" ($y_1 = 16,000$), "white-collar" ($y_2 = 80$), and "blue-collar" ($y_3 = 15$).
   - The first state provides the opportunity and incentive for a high concentration of earnings and wealth (see Castañeda, Diaz-Gimenez, and Rios-Rull, 2003). $\Gamma_{2,1} = 0.001$, $\Gamma_{1,1} = 0.02$.
   - The persistence of the latter two states ensures that earnings display a sizable positive autocorrelation: $\Gamma_{2,2} = 0.78$, $\Gamma_{3,3} = 0.964$.
   - This specification requires 6 parameters in the Markov transition matrix but we reduce it to 4 by setting the probability of moving from blue-collar to super-rich and vice-versa to zero. $\Gamma_{3,1} = 0$, $\Gamma_{1,3} = 0$, $\Gamma_{1,2} = 0.98$, $\Gamma_{2,3} = 0.219$, $\Gamma_{3,2} = 0.036$.
   - Let's assume agents cannot borrow, $\underline{a} = 0$.

   (a) (20 points) Solve this problem using discretization and policy function iteration with 30 and 500 points in the grid for a with the same range. Report how much time each case takes. Plot $a'(a, y_2)$ with a in the x axis for the two cases in the same figure to compare the results. (include a 45 degree line, too)

(b) (10 points) Plot $a'(a, y_1)$ for the case of 500 grid points (include a 45 degree line). Then, plot $a'(a, y_3)$ for the same case (include a 45 degree line, too). What are the economic forces that explain the difference in the saving behavior of "super-rich" and "blue-collar" workers?

(c) (20 points) Now solve the problem using 30 points in the grid and Chebyshev interpolation using polynomial of order N=4. Report how much time it takes. Repeat the plots in part (a) with this new solution and the solution using discretization with 500 points.

(d) (20 points) Now solve the problem using 30 points in the grid and the endogenous grid point method. Report how much time it takes. Repeat the plots in part (a) with this new solution and the solution using discretization with 500 points.

(e) (15 points) Pick the method that worked better for you (What is it?) Solve the problem using that method but now setting $\underline{a} = -30$. Plot $a'(a, y_3)$ for this case and the case with $\underline{a} = 0$ in the same figure. Explain how and why the saving behavior of "blue-collar" workers change.

(f) (15 points) Again, using your preferred method, solve the problem but now setting $\sigma = 10$. Plot $a'(a, y_1)$, $a'(a, y_2)$, and $a'(a, y_3)$ for this case and the case with $\sigma = 2$ in the same figure (3 plots in total). Explain how and why the saving behavior of workers change.